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Anatomical language

Course notes intended for first-year medical students.

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ANATOMICAL LANGUAGE

Course outline

Introduction

1. Anatomical language
2. Dynamic benchmarks
3. Location and direction terms
4. Structural organization of the human body
5. Topographical organization of the human body

Conclusion

Bibliography

Introduction

Anatomical language, or anatomical nomenclature, is a International Nomenclature of Human Anatomy, written in Latin, the result of an international agreement, aims to universalize the study of the human body.

1. Anatomical language

1.1. Nomina Anatomica (NA)

For a "universal" study of the human body, an international agreement (adopted in Paris in 1955) developed an international anatomical nomenclature in Latin, the Nomina Anatomica: NA

This nomenclature relates to a well-defined position of the human body, where 3 axes are distinguished, which allow us to define 3 planes, and reference points which designate the direction of movements.

1.2. Reference anatomical position

All anatomical descriptions refer to a "Reference Anatomical Position", where the subject is:

- A living, upright human body.
- The upper limbs hang down along the trunk.
- Palms facing forward.
- The gaze should be straight and horizontal.
- With legs extended,
- Feet together, placed on the ground

1.3. Reference anatomical axes

1.3.1. The body's axis: It is the vertical line dropped from a point located at the top of the skull that passes through the center of gravity of the body (located at the level of the pelvis).

1.3.2. The axis of the hand : corresponds to the longitudinal axis that passes through the 3rd finger.

1.3.3. The axis of the foot : corresponds to the longitudinal axis that passes through the 2nd toe.

1.4. Reference plans

Three perpendicular planes:

- **The frontal (coronal) plane:**vertical plane parallel to the forehead, which divides the body into 2 parts:
  - Previous.
  - Posterior.
- **The sagittal plane:**vertical plane, anteroposterior. It can be:
  - Median sagittal: passes through the midline and divides the body into 2 symmetrical parts, right and left.
  - Sagittal paramedian: passes beside the midline and divides the body into 2 parts, internal and external.
- **The transverse plane (horizontal or axial):**a plane parallel to the ground, which divides the body into 2 parts:
  - Superior.
  - Lower.

2. Dynamic benchmarks

Body movements occur in a plane, and in relation to an axis.

- **Along the transverse axis**we have:

Flexion: brings a part of the body forward

Extension: brings a part of the body backwards.

- **Along the vertical axis**we have:

Internal rotation: rotary movement from outside to inside.

External rotation: rotary movement from the inside out.

- **Along the sagittal axis**we have:

Adduction: brings a part of the body closer to the midline.
Abduction: moves a part of the body away from the midline.

3. Location and direction terms

- **Medial:**near the median sagittal plane.
- **Lateral:**far from the median sagittal plane.
- **Superior, cranial, or cephalic:**near the upper end.
- **Inferior or caudal:**near the lower end.
- **Proximal:**close to the root of the limb.
- **Distal:**far from the root of the limb.
- **Right :**closer to the right.
- **LEFT :**closer to the left.
- **Ipsilateral:**belong to the same side.
- **Contralateral:**belong to two different sides.
- **Anterior or ventral:**located near the anterior wall.
- **Posterior or dorsal:**located near the posterior wall.
- **Internal :**look inside a cavity or organ.
- **External :**look at the outside of a cavity or organ.
- **Superficial:**close to the surface.
- **Deep :**far from the surface.

4. Structural organization of the human body

4.1. Levels of structural organization

- **The chemical level**: it is the set of chemical substances necessary for the maintenance of life.
- **The cellular level**: the cell is the basic structural and functional unit.
  Ex: neuron, blood cell.
- **The tissue level**: tissue is a group of cells that play a homologous role.
  Example: bone tissue.
- **The organic level**: the organ brings together different tissues for a specific function.
  Examples: liver, brain.
- **The systemic level**: the system is a set of organs that are comparable from a structural point of view.
  Ex: skeletal system, central nervous system.
- **The systemic level**: the apparatus is the set of dissimilar organs which participate in the same function.
  Examples: digestive system, cardiovascular system.

4.2. Organ systems

4.2.1. Devices of relational life

4.2.1.1.The locomotor system: encompasses the skeletal, articular and muscular systems.
4.2.1.2. The nervous system: includes the central, peripheral and autonomic nervous systems.
4.2.1.3. The sensory apparatus: includes the 5 sense organs (touch, taste, smell, sight and hearing).

4.2.2. Nutrition devices

4.2.2.1.The digestive system:includes the digestive tract and associated glands.
4.2.2.2.The circulatory system:includes the heart and the arterial, venous and lymphatic systems.
4.2.2.3.The respiratory system:includes the respiratory tract and lungs.
4.2.2.4.The urinary system:includes the kidneys and urinary tract.
4.2.2.5.The endocrine system:represented by the endocrine glands.

4.2.3. Generating (or reproducing) apparatus

4.2.3.1.Male genitalia: includes the testicles, prostate, and spermatic ducts
4.2.3.2.Female genitalia: includes the ovaries and the genital tract.

5. Topographical organization of the human body

5.1.Human body shape

The shape of the human body is determined by that of the osteo-cartilaginous skeleton. Muscles are added to fill and reinforce the walls.

5.2. The head

It carries the sensory apparatus. It contains the initial segments of the digestive and respiratory systems, and it contains a large part of the nervous system.

5.3.The neck

It supports the head. It is a passageway for visceral, vascular, and nervous elements.

5.4.The trunk

Contains the major cavities of the body: The thoracic cavity, at the top, and the abdominopelvic cavity, at the bottom.

5.5.The members

The limbs are extensions of the trunk, formed essentially by bones, joints, muscles, vessels and nerves.

CONCLUSION

Anatomy is an ancient and fundamental science, essential for understanding physiology and mastering radiology and surgery. Effective and uncomplicated surgical interventions are based on a thorough understanding of anatomy.

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